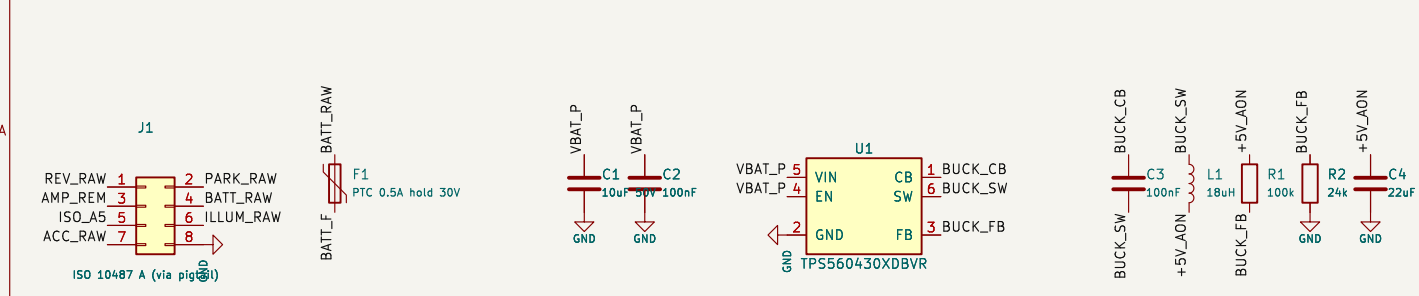
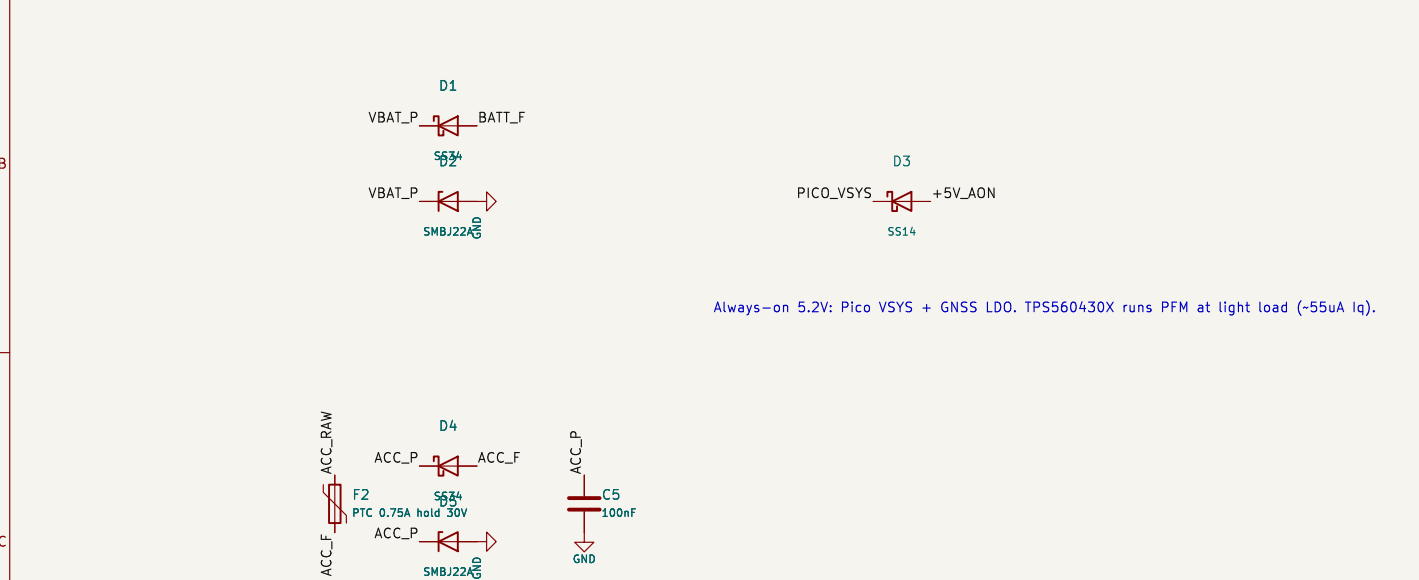


1. ISO 10487 CONNECTOR A (power) + ALWAYS-ON 5V FOR PICO AND GNSS

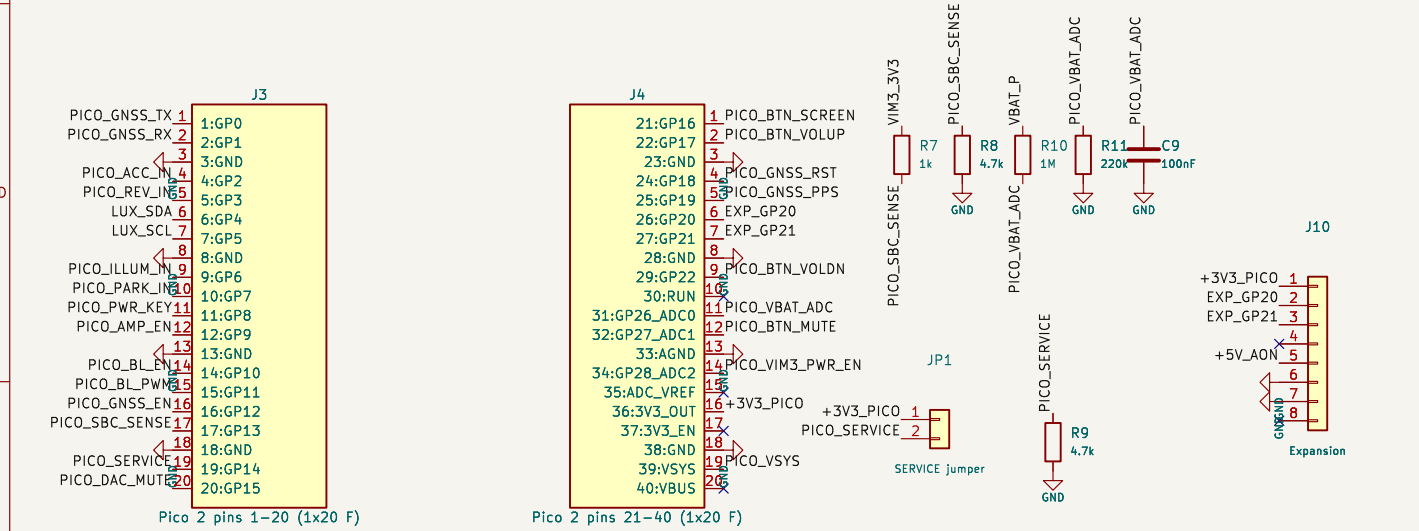


J1 pad n = ISD A pin n. A1 REVERSE A2 HANDBRAKE A3 AMP REMOTE A4 BATT+ A5 remote (R57) A6 ILLUM A7 ACC A8 GND



Always-on 5.2V: Pico VSYS + GNSS LDO. TPS560430X runs PFM at light load (~55uA Iq).

4. PICO 2 (on two 1x20 sockets) + USB to VIM3



USB: Pico micro-USB cable to a VIM3 USB-A port (HID vehicle data + CDC GNSS bridge).

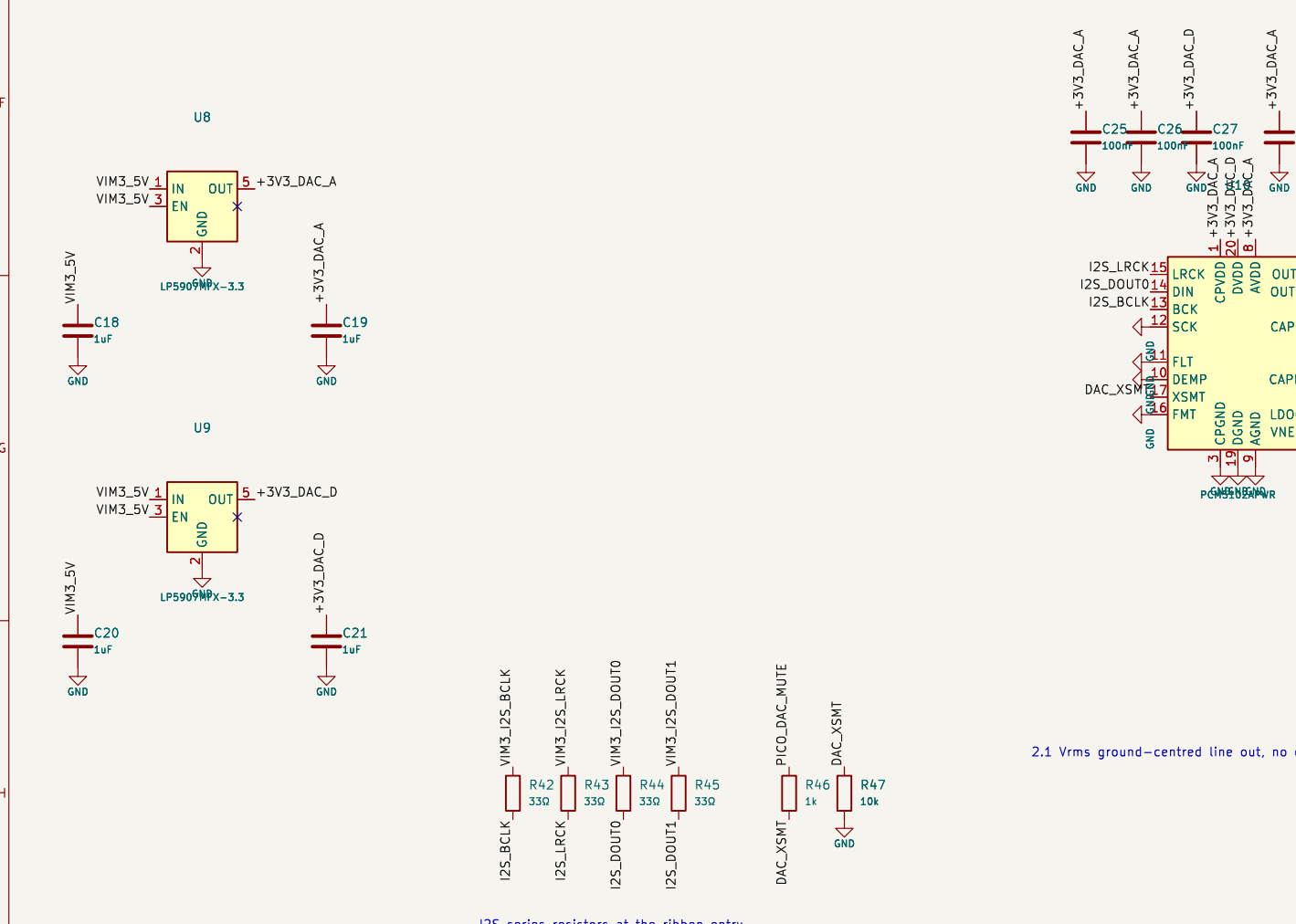
CBD-14, CNEC-WATER, CBD-100, CBT-BEN, CBI-15, CBG-LIN, CBI-ILLUM, COT-DARK, CBO-DWD

GP0/1 GNSS DART0, GP2 ACC, GP3 REV, GP4/5 I2CO IUX, GP6 ILLUN, GP7 PARK, GP8 PWR key, GP9 AMP.

GP10 BL_EN, GP11 BL_PWM, GP12 GNSS_EN, GP13 SBC sense, GP14 service, GP15 DAC mute, GP18 GNSS reset

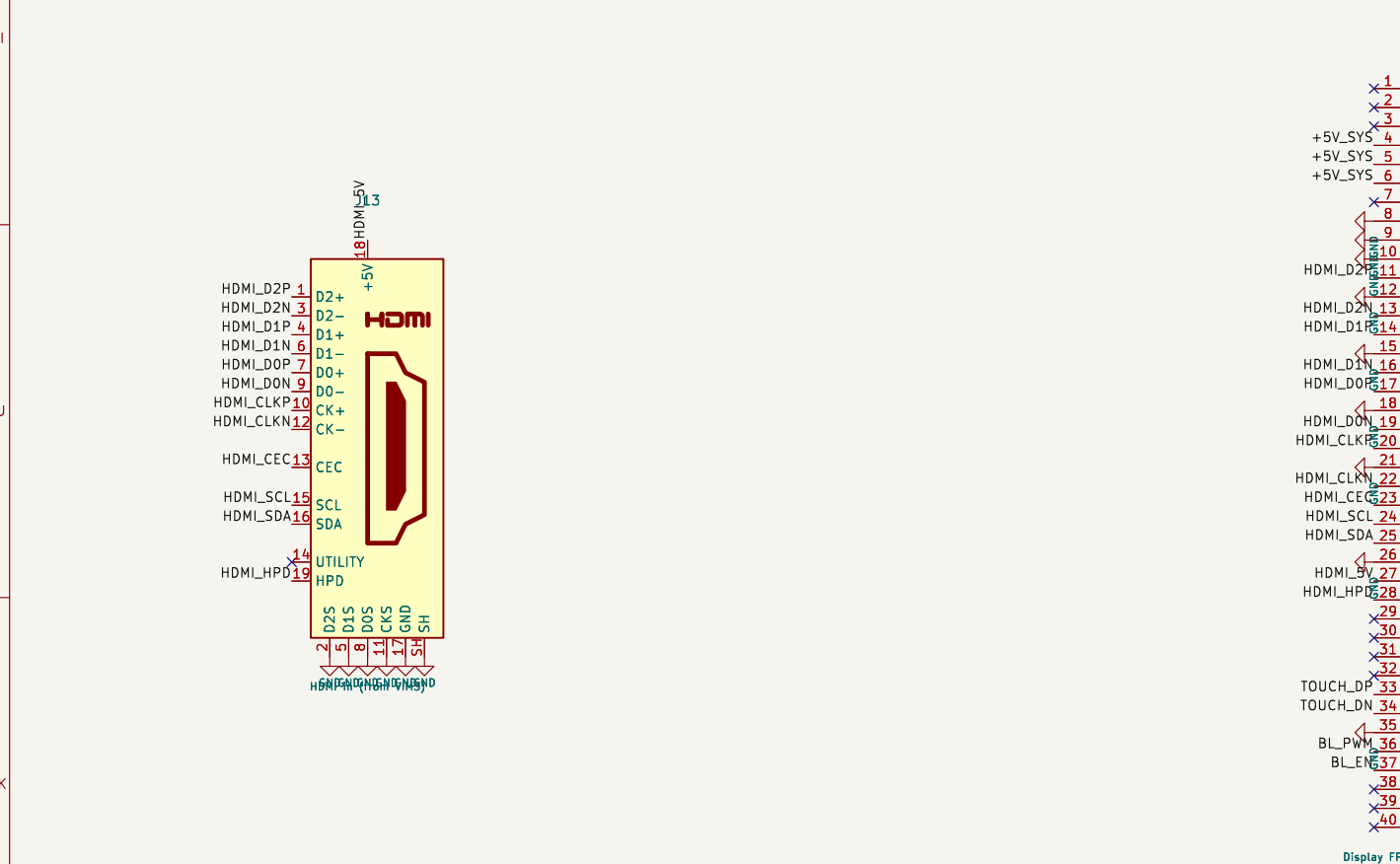
GP19 GNSS PPS, GP20/21 UART1 (K-line, reserved), GP16/17/22/27 display buttons, GP28 VIN3 5V supply e

7. AUDIO: VIM3 TDM-B I2S -> 2x PCM5102A (front + rear) -> line out on ISO C1



Car radio peripheral board rev 0.2 - MG F (1996), VIM3 + Pico 2, 4ch line out, NEO-M9N. Generated; see DESIGN.md.

9. DISPLAY: Waveshare 7in 70H-1024600 adapter built in (HDMI in from VIM3 -> 40-pin FPC to the panel)

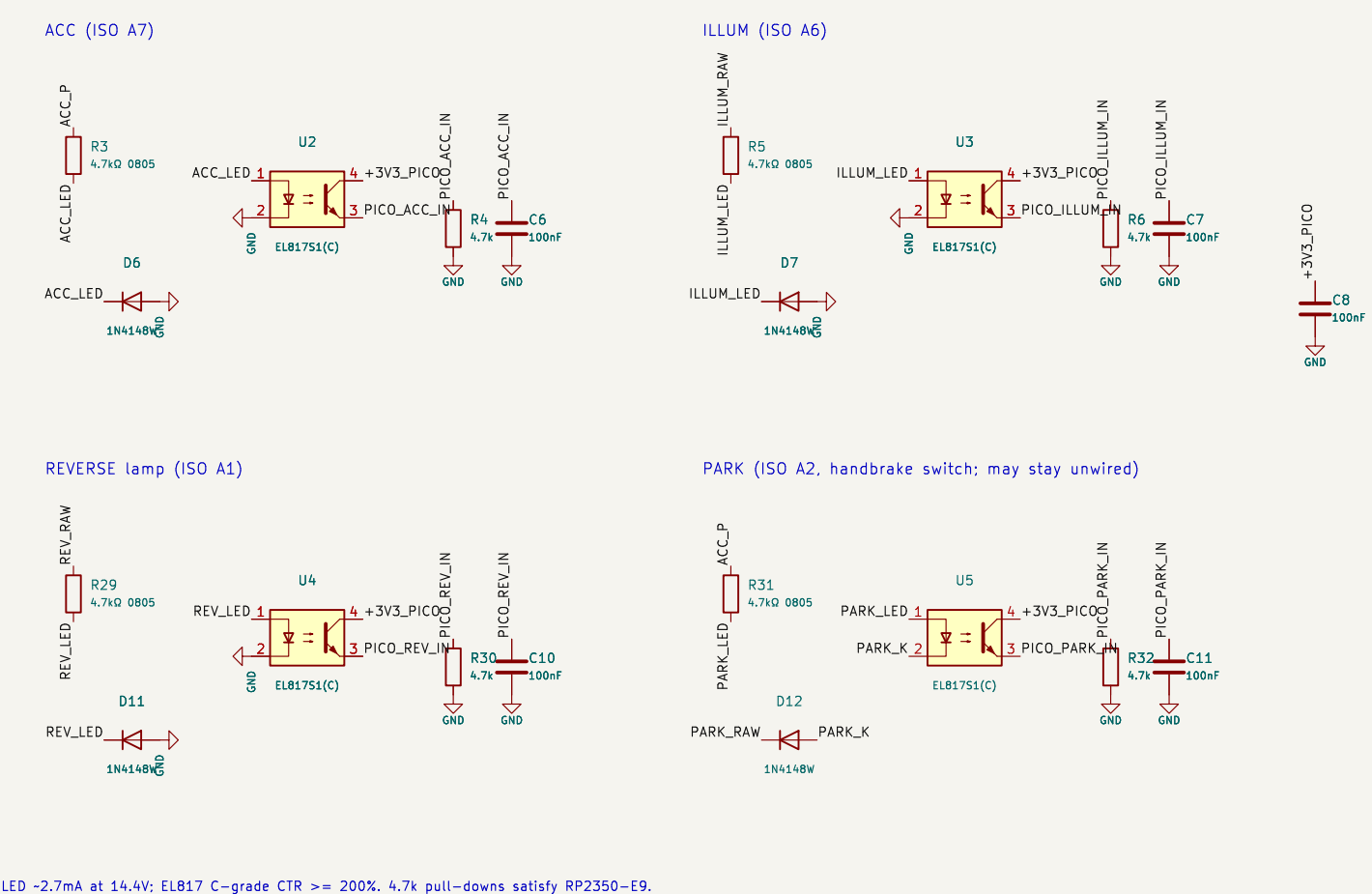


Panel 5V (+0.45A) comes from the board's switched 5V rail (+5V_SYS, section 10), not from VIM3 USB. Touch USB goes to the on-board hub (section 11).

Backlight EN/PWM now go straight to the panel; 0V = brightest, 3.3V = darkest (invert PWM in firmware). J6 removed.

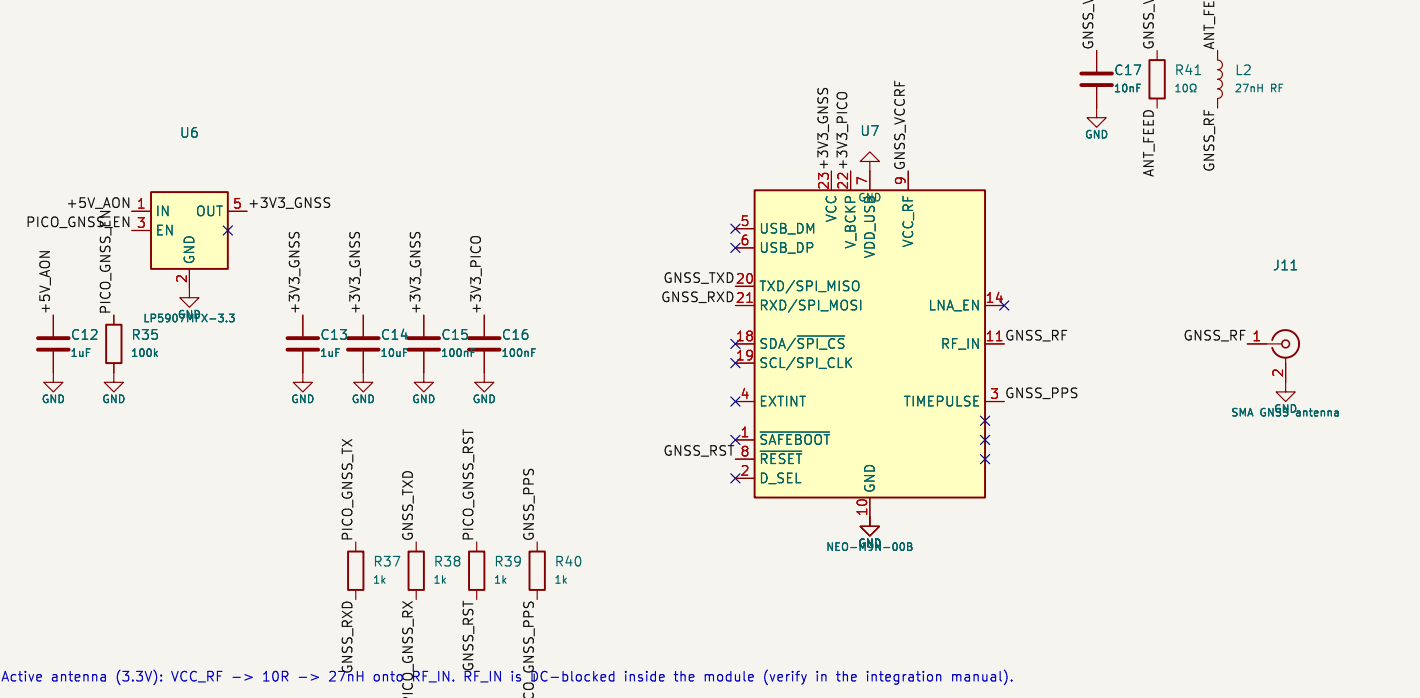
FPC orientation as on the Waveshare adapter: seen from the top, ribbon leaving the board edge, pin 1 is on the RIGHT, pin 40 on the left.

2. VEHICLE INPUTS (opto-isolated). ACC, ILLUM, REVERSE: +12V = on. PARK: switch to ground = on.



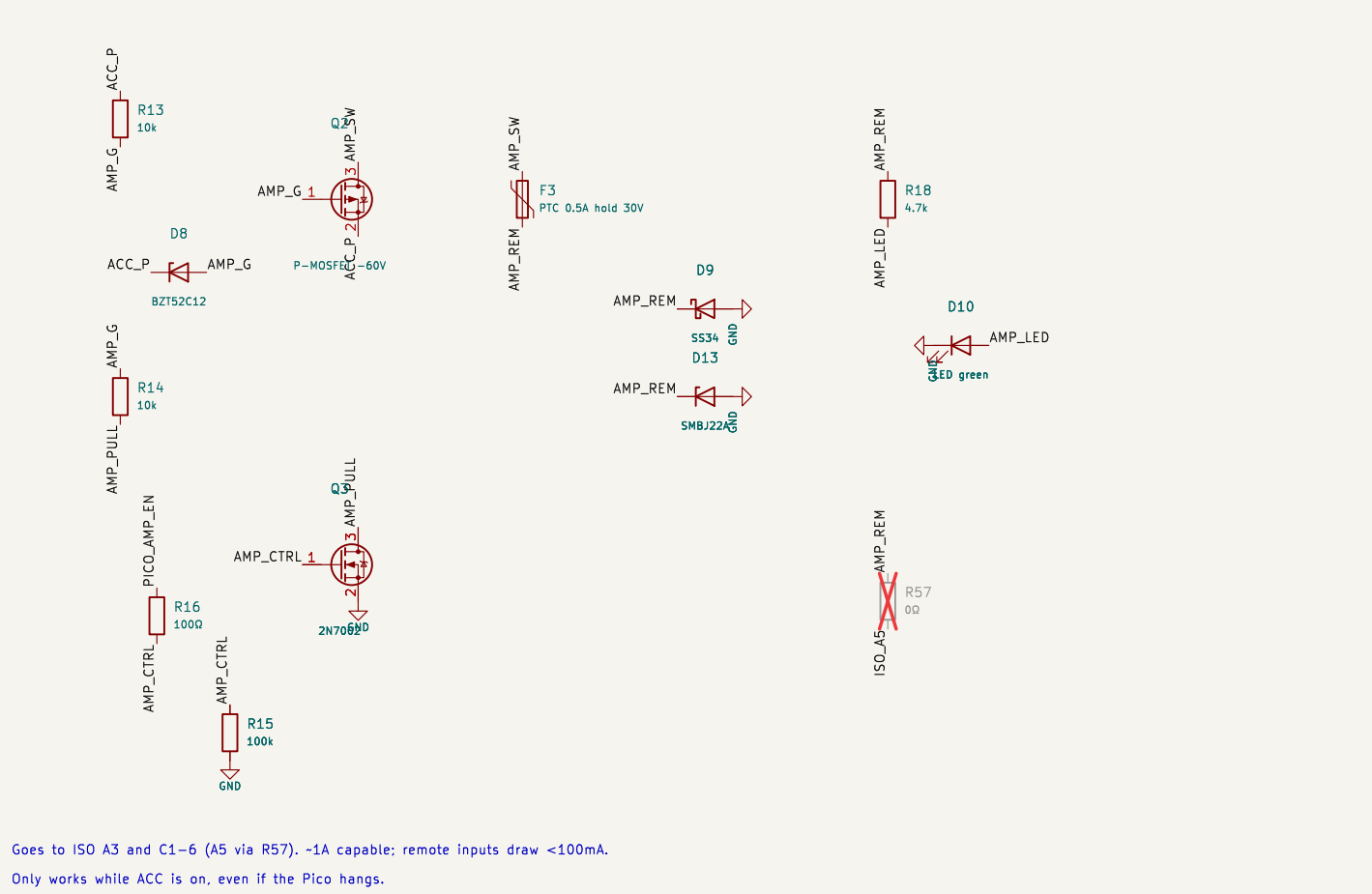
LED -2.7mA at 14.4V; EL817 C-grade CTR $\geq 200\%$. 4.7k pull-downs satisfy RP2350-E9

5. GNSS: u-blox NEO-M9N, active antenna on SMA



Active antenna (3.3V): VCC_RF → 10R → 27nH onto RF_IN. RF_IN is DC-blocked inside the module (verify in the integration manual).

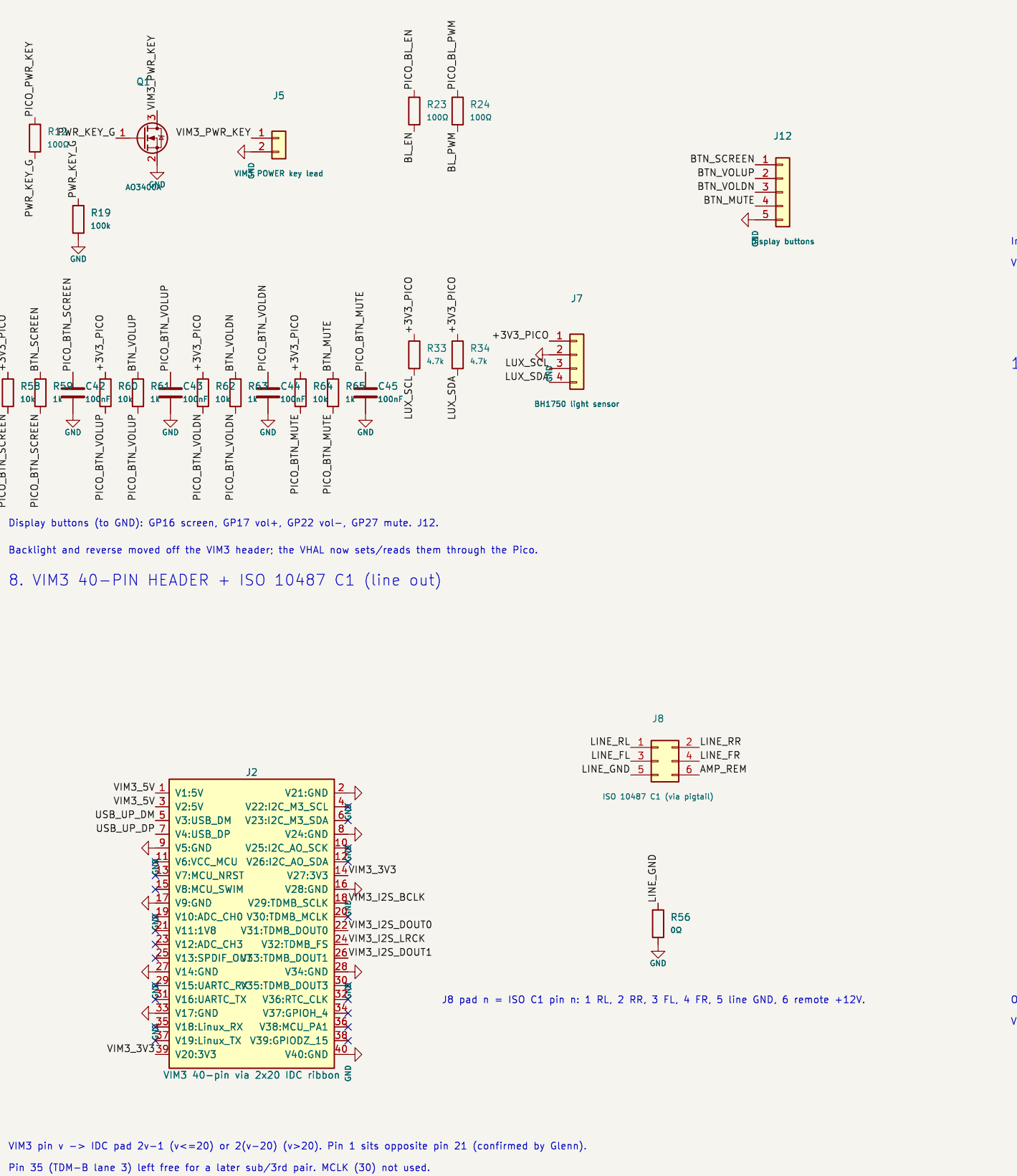
3. AMPLIFIER REMOTE (+12V from ACC, Pico GP9)



Goes to ISO A3 and C1-6 (A5 via R57). ~1A capable; remote inputs draw <100mA

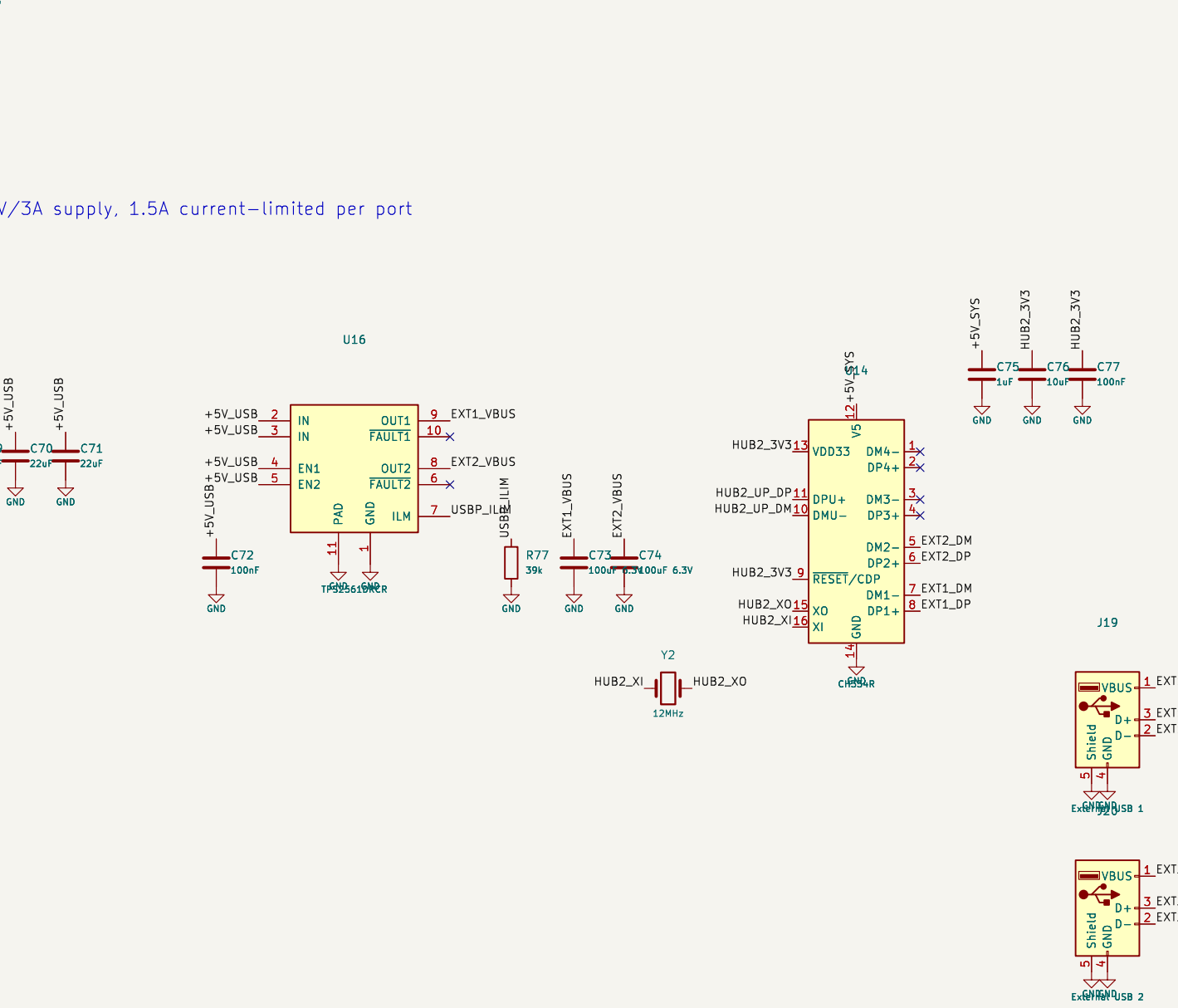
Only works while ACC is on, even if the Pico hangs

6. VIM3 POWER KEY, BACKLIGHT, LIGHT SENSOR (all from the Pico)

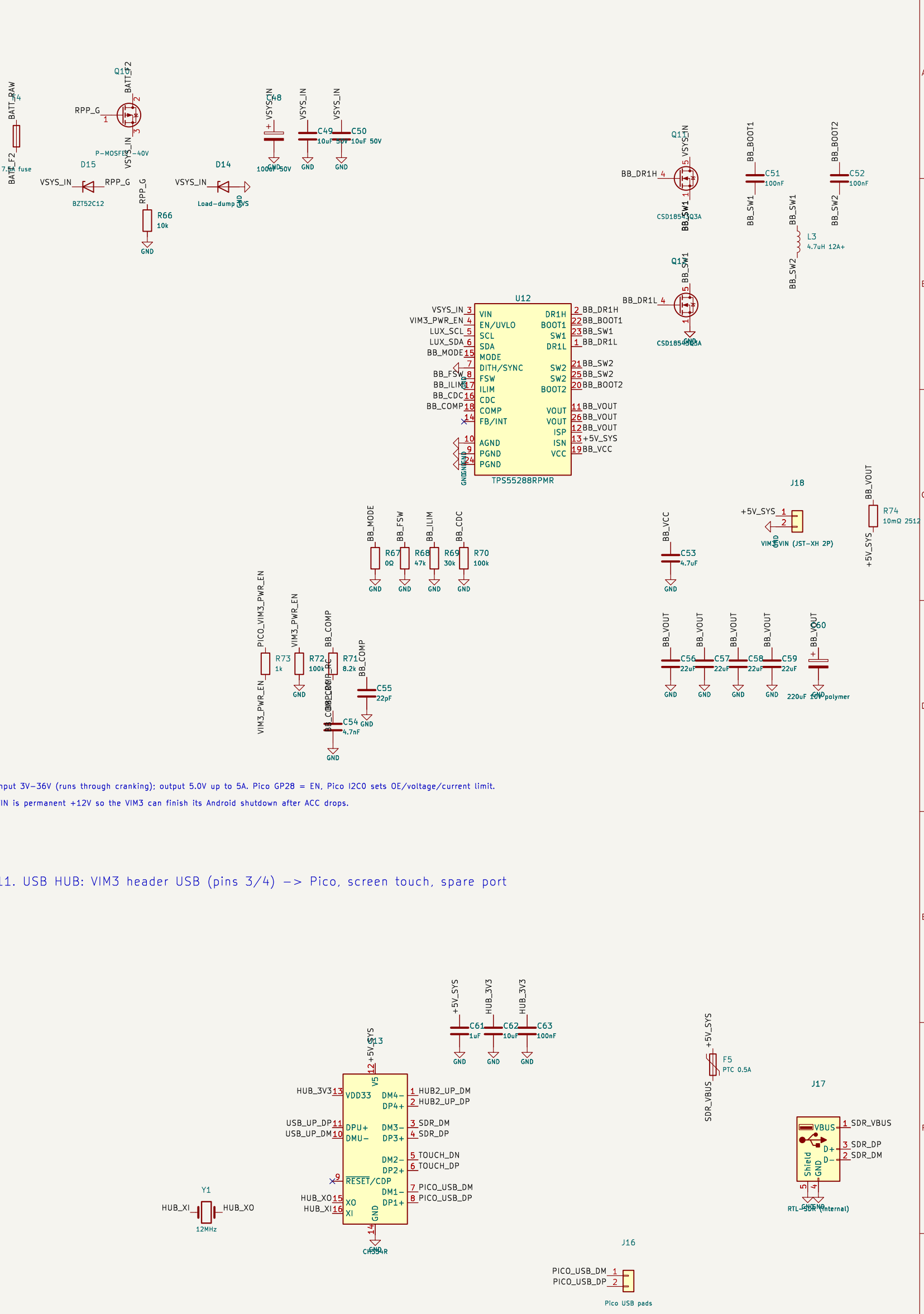


VIM3 pin v => IDC pad 2v-1 (v<=20) or 2(v-20) (v>20). Pin 1 sits opposite pin 21 (confirmed by Glenn)

Pin 35 (TDM-B lane 3) left free for a later sub/3rd pair. MCLK (30) not used.



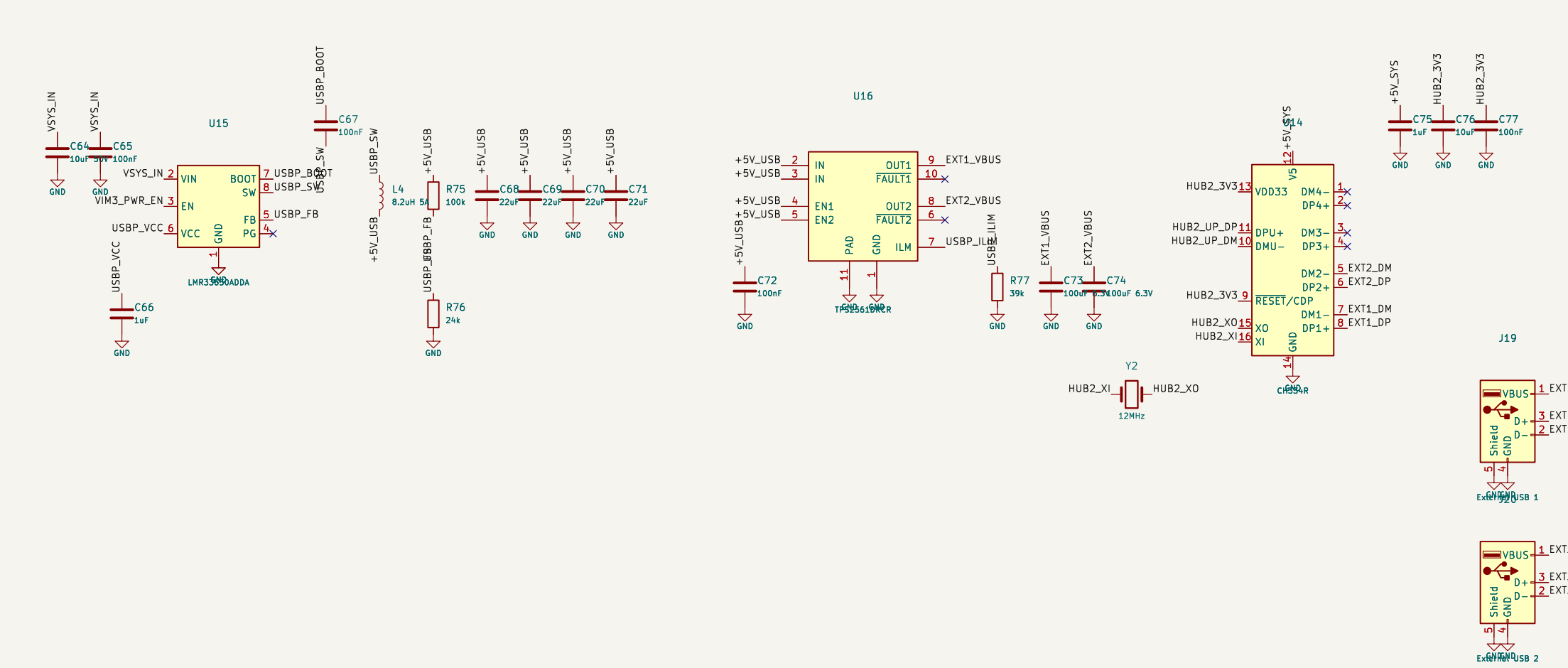
0. 5V/5A SYSTEM SUPPLY FOR VIM3 + SCREEN + HUB (TPS55288 buck-boost, rides through cranking and overvoltage)



line link to the VIN3: no USB cables. Port 1 Pico (HID + GNSS CDC), 2 touch, 3 RTL-SDR, 4 second hub (external ports). Route D+/D- as 90 ohm pairs

IM3 header pins 3/4 are a working USB host port (confirmed by Glenn, 2026-09-29)

12. EXTERNAL USB PORTS (phone etc.): second hub + dedicated 5V/3A supply, 1.5A current-limited per port



Ports are powered only while the VIM3 supply is on. All USB shares the VIM3 header's USB 2.0 link (480 Mbit/s); RTL-SDR needs ~40 Mbit/s